

OpenShift 5-Day Training

Total Questions: 25

Difficulty Level: Hard

Question Type: Scenario-based MCQ

Recommended Duration: 50–60 minutes

Coverage: Day 1 to Day 5 full OpenShift training

Section A: Scenario-Based MCQs

1. VM vs Container vs OpenShift

A banking client currently deploys each application on separate VMs. The platform team wants faster deployment, better resource utilization, and automated recovery if an application crashes.

Which explanation best justifies OpenShift adoption?

- A. OpenShift removes the need for an operating system.
 - B. OpenShift replaces all VMs with physical servers.
 - C. OpenShift packages applications as containers and manages their deployment, scaling, networking, security, and recovery.
 - D. OpenShift is mainly used to create more VMs automatically.
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2. Container Image Lifecycle

A developer commits code to Git. OpenShift builds the application using S2I, creates an image, stores it internally, and deploys the application.

Which sequence best represents the lifecycle?

- A. Source Code → Running Pod → ImageStream → BuildConfig → Registry
 - B. Source Code → BuildConfig/S2I → Image → Internal Registry/ImageStream → Deployment → Pod
 - C. ImageStream → Source Code → Pod → BuildConfig → Registry
 - D. Pod → BuildConfig → Source Code → ImageStream → Route
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3. External Registry vs Internal Registry

A team deploys `nginx` directly from Docker Hub using an image reference. Later, they assume the image is automatically copied to OpenShift's internal registry.

What is the correct understanding?

- A. Every external image is automatically copied to the internal registry.
 - B. OpenShift cannot pull images from Docker Hub.
 - C. The image is pulled from the external registry unless it is imported, built, mirrored, or promoted into the internal registry.
 - D. Docker Hub images can only run on Kubernetes, not OpenShift.
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4. OverlayFS Layering Issue

A company uses a large base image containing OS libraries, Java runtime, and application code in the same layer. Each small code change causes large image rebuilds and slow pulls.

What is the best improvement?

- A. Put frequently changing application code in the lower image layer.
 - B. Separate stable base dependencies from frequently changing application code so only changed layers are rebuilt and pulled.
 - C. Remove all image layers and build one flat image.
 - D. Store the application directly inside a running pod.
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5. Pod Failure Scenario

A production application pod crashes because of a memory issue. The platform automatically creates a replacement pod, but the new pod gets a different IP address.

Which object ensures users do not depend on changing pod IPs?

- A. ConfigMap
 - B. Secret
 - C. Service
 - D. BuildConfig
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6. Namespace vs Project

A Kubernetes engineer says:

“OpenShift Project is just a Namespace, so there is no practical difference.”

Which response is most accurate?

- A. A Project is completely unrelated to Namespace.
 - B. A Project is a Kubernetes Namespace with OpenShift-added management, access control, quota, and developer experience features.
 - C. Namespace is only for storage, while Project is only for networking.
 - D. Project exists only in the OpenShift web console, not in CLI.
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7. YAML Deployment Validation

A learner creates a Deployment YAML file but misses the `selector` and pod template label alignment. The deployment is created but does not manage the expected pods correctly.

What is the most likely root cause?

- A. The container image is too large.
 - B. The Deployment selector does not match the pod template labels.
 - C. The Service type should be NodePort.
 - D. The ConfigMap is missing.
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8. OpenShift Architecture

A company wants to separate application workloads from cluster infrastructure components such as routers, registry, and monitoring.

Which node design is most suitable?

- A. Run all workloads only on control plane nodes.
 - B. Use worker nodes for applications and infra nodes for infrastructure components.
 - C. Use only one node for all workloads.
 - D. Use external registry nodes as worker nodes.
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9. Operators and OLM

A team wants to deploy a database with automated upgrades, backup configuration, and lifecycle management. They do not want to manually maintain all YAML manifests.

Which OpenShift capability is most appropriate?

- A. Route
- B. Operator managed through OLM
- C. ConfigMap
- D. DeploymentConfig only

10. `oc` VS `kubectl`

A Kubernetes admin can run `kubectl get pods` successfully in OpenShift but cannot easily create OpenShift-specific resources such as Routes or start builds.

Which tool should they use and why?

- A. Use `oc` because it supports Kubernetes resources plus OpenShift-specific resources and workflows.
 - B. Use only `kubectl` because `oc` is deprecated.
 - C. Use Docker CLI because OpenShift does not support Kubernetes CLI commands.
 - D. Use Helm only because OpenShift does not support CLI-based deployment.
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11. RBAC Least Privilege

A developer needs to deploy and update applications in only one project. They should not manage cluster-level resources or permissions across other projects.

Which role assignment is most appropriate?

- A. cluster-admin at cluster level
 - B. edit role at project level
 - C. view role at cluster level
 - D. self-provisioner role only
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12. Service Account Security

An application needs to read ConfigMaps and Secrets at runtime. A developer suggests mounting their own personal OpenShift token inside the pod.

What is the best security response?

- A. Use the developer token because it is easier.
 - B. Use a dedicated ServiceAccount with minimum required RBAC permissions.
 - C. Use cluster-admin token to avoid permission issues.
 - D. Store the token in a ConfigMap.
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13. S2I Use Case

A Java developer has source code in Git but does not know Dockerfile syntax. The organization wants a standard build process controlled by the platform team.

Which OpenShift feature is most suitable?

- A. Source-to-Image
 - B. PersistentVolume
 - C. Route
 - D. NodePort
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14. BuildConfig Trigger

A team wants a new build to start automatically whenever source code changes in Git. They also want deployment to happen when a new image is available.

Which combination is most appropriate?

- A. Git webhook trigger and ImageChange trigger
 - B. Manual pod deletion and Service restart
 - C. Route trigger and PVC trigger
 - D. Secret trigger and NodePort trigger
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15. ImageStream Promotion

An image is built in the `dev` project and tested successfully. The same image should be promoted to `test` without rebuilding from source.

What is the best OpenShift-native approach?

- A. Rebuild the image separately in test from the same Git branch.
 - B. Use ImageStream tagging/import/promotion to reference the tested image in the next environment.
 - C. Copy the running pod from dev to test.
 - D. Export the pod logs and import them into test.
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16. DeploymentConfig vs Deployment

An older OpenShift application uses ImageChange triggers and automatically redeploys whenever a new ImageStream tag is updated.

Which workload object is most commonly associated with this behavior?

- A. Kubernetes Deployment
 - B. DeploymentConfig
 - C. PersistentVolumeClaim
 - D. Route
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17. Rollback Scenario

After a new application version is deployed, users report errors. The previous version was stable. The team wants to restore the earlier version quickly.

Which OpenShift capability helps?

- A. Rollback to a previous deployment revision
 - B. Delete the project permanently
 - C. Remove the Service
 - D. Change ClusterIP manually
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18. ConfigMap Update Behavior

An application receives configuration through environment variables from a ConfigMap. The ConfigMap is updated, but the running pod still uses the old value.

Why?

- A. ConfigMaps cannot be updated.
 - B. Environment variables are injected at container startup and usually require pod restart to reflect changes.
 - C. ConfigMaps work only with Secrets.
 - D. OpenShift does not allow ConfigMaps in production.
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19. Secret Handling

A team stores database passwords in plain text inside deployment YAML files committed to Git.

What is the biggest issue?

- A. YAML files cannot contain environment variables.
- B. Passwords should not be stored in plain text Git repositories; Secrets or external secret management should be used.
- C. Passwords must always be stored in ConfigMaps.
- D. OpenShift does not support database passwords.

20. Route TLS Termination

A security team wants HTTPS traffic from the user to OpenShift router and HTTPS again from router to backend pod. The router should decrypt and then re-encrypt traffic.

Which TLS termination type should be used?

- A. Edge
- B. Re-encrypt
- C. Passthrough
- D. NodePort

21. Canary Deployment

A team wants to release version 2 of an application only to 10% of users while 90% traffic continues to version 1.

Which strategy best fits?

- A. Recreate deployment
- B. Canary deployment
- C. Delete old version immediately
- D. Scale database to zero

22. Persistent Storage

A PostgreSQL pod stores data inside the container filesystem. After pod recreation, all data is lost.

Which design change is required?

- A. Use a Route
- B. Use a PersistentVolumeClaim backed by a PersistentVolume or dynamic StorageClass
- C. Use a ConfigMap for database files
- D. Use a ServiceAccount

23. Security Context Constraints

A vendor container requires root privileges. OpenShift blocks the pod from running due to default security restrictions.

Which is the correct production-grade response?

- A. Give all users cluster-admin access.
 - B. Modify the application image to run as non-root wherever possible; if root is unavoidable, assign a suitable SCC only to a dedicated ServiceAccount after risk review.
 - C. Disable OpenShift security completely.
 - D. Run the container directly on the node outside OpenShift.
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24. HPA and Resource Requests

A team configures Horizontal Pod Autoscaler based on CPU utilization, but autoscaling does not work properly. On investigation, the deployment has no CPU requests defined.

What is the likely issue?

- A. HPA usually needs CPU requests to calculate utilization percentage correctly.
 - B. HPA works only with Routes.
 - C. HPA requires PersistentVolumeClaims.
 - D. HPA works only with DeploymentConfig, not Deployment.
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25. Capstone Architecture Scenario

A final project requires deploying a backend API with database, TLS exposure, CI/CD pipeline, autoscaling, and persistent storage.

Which architecture is most appropriate?

- A. Single pod containing API, database, CI/CD engine, and storage inside container filesystem.
 - B. Separate API Deployment, database with PVC, Service for internal access, Route with TLS for external access, pipeline for build-test-deploy, and HPA for API scaling.
 - C. Static pod with no Service, no Route, and no storage.
 - D. Manual deployment from developer laptop with no pipeline.
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Answer Key

Q.No Answer

- 1 C
- 2 B
- 3 C
- 4 B

Q.No Answer

5	C
6	B
7	B
8	B
9	B
10	A
11	B
12	B
13	A
14	A
15	B
16	B
17	A
18	B
19	B
20	B
21	B
22	B
23	B
24	A
25	B